THESIS RESEARCH GROUND

Canonical Thesis Field Specification

Multimodal / Vision-Language Learning and Systems

One frozen parent field. Three ranked research fields. Topic discovery remains active.

Version	1.0
Status	Field Frozen | Topic Not Frozen
Effective date	22 August 2026
Use	Private canonical reference for thesis research and publication planning

This document governs field and topic decisions. It is not a university form, schedule, or promise of publication.

Executive Field Decision

CANONICAL DECISION The frozen parent field is Multimodal / Vision-Language Learning and Systems. The ranked research fields are: (1) Trustworthy and Reliable VLMs, (2) VLM Generalization, Adaptation and Distribution Shift, and (3) Efficient and Scalable VLMs. A thesis topic has not been selected.

- **Primary research field:** Trustworthy and Reliable VLMs.
- **Secondary research field:** VLM Generalization, Adaptation and Distribution Shift.
- **Conditional research field:** Efficient and Scalable VLMs.
- **Publication strategy:** design the work to publication standards from the beginning, complete a defensible thesis first, then submit or extend only when the evidence is mature.
- **Operating rule:** no unrelated field may be reopened without a written change request and evidence that the frozen field has become infeasible.

How to Use This Document

Use this specification before starting a literature review, creating a candidate topic, selecting a dataset, choosing a model, or accepting a claimed research gap. Every decision-changing research session must end with a verdict, evidence, limitation, next action, and decision-log entry.

- Sections 1–4 define the field, ranked research directions, and boundaries.
- Sections 5–8 govern topic discovery, feasibility, contribution quality, and experiments.
- Sections 9–12 govern compute, data, reproducibility, publication, and workflow.
- Sections 13–15 provide templates, checklists, and change control.

Document Authority and Revision Rules

This is the single source of truth for personal thesis research. ChatGPT, Claude, Perplexity, Kimi, notebooks, and temporary notes are workbenches. A suggestion from any platform is provisional until it is checked against this document and recorded in the Decision Log.

- Field changes require a major version and evidence that the current parent field cannot produce a feasible topic.
- Research-lane ranking changes require a minor version and a written evidence update.
- A topic is not frozen until it passes the Topic Feasibility Gate and the Topic Freeze Record is completed.
- Dates and university administration are intentionally excluded because this is a private research specification.

1. Researcher Profile and Operating Constraints

The specification is calibrated for a final-year undergraduate CSE researcher in Bangladesh whose primary career direction is AI/ML engineering and research. The research plan assumes substantial self-direction and does not count on continuous supervisor involvement.

Dimension	Canonical position
Academic foundation	Algorithms, Pattern Recognition, Artificial Intelligence, Computer Networks, Object-Oriented Programming, and Operating Systems.
Technical foundation	Classical machine learning, deep learning, CNNs, transformers and LLM basics, Python, scikit-learn, PyTorch/TensorFlow familiarity, and web/API development.
Career objective	Build credible AI/ML research ability and end-to-end engineering evidence for employment, while preserving a path to funded graduate study.
Research team	Three members, with the principal researcher expected to lead most research design, implementation, evaluation, and writing.
Supervision assumption	Treat the thesis as largely self-directed. External feedback is useful but not a feasibility dependency.
Local compute	Ryzen 5 5600G, 8 GB RAM, integrated graphics. Local work is for coding, small preprocessing, documentation, and lightweight tests.
GPU compute	Google Colab Pro is the normal GPU environment. Limited temporary access to stronger GPUs may be used only after a Colab-feasible pilot succeeds.
Financial boundary	No dependence on expensive model APIs, private datasets, custom hardware, large cloud bills, or paid annotation.
Publication objective	Publication-aware design from day one; completion of a strong thesis takes priority over forcing an early submission.

1.1 Non-negotiable Research Properties

- The core contribution must remain AI/CSE-heavy and understandable without unavailable domain experts.
- Data and model weights must be publicly accessible or obtainable under clear academic-use terms.
- A strong baseline must be reproducible before a new method is proposed.
- The main claim must be tested by direct metrics, ablations, and error analysis.
- The minimum experiment must fit Colab Pro and remain repeatable across seeds or controlled runs.
- The contribution must be more than applying a known model to a new dataset.

HARD TRUTH A fashionable field does not create a publishable thesis. Publication depends on a defensible question, credible evidence, controlled experiments, and honest limitations. Venue acceptance is never treated as guaranteed.

2. Frozen Parent Field

2.1 Canonical Definition

Multimodal / Vision-Language Learning and Systems studies models and evaluation methods that connect visual information with natural language for perception, grounding, retrieval, reasoning, generation, and decision support. In this specification, the field includes model behavior, evaluation, adaptation, and deployment efficiency. It excludes training frontier foundation models from scratch.

2.2 Evidence for the Field Lock

The field lock is based on fit and continuity, not on the ordering of an AI-generated topic list. The official CVPR 2025 and CVPR 2026 calls both include multimodal learning, vision-language reasoning, datasets and evaluation, efficient and scalable vision, and transfer or continual learning [1, 2]. CVPR 2026 also scheduled dedicated oral sessions for multimodal vision and efficient reasoning, providing additional evidence of active research attention [3].

Decision factor	Assessment	Consequence
Background fit	Strong	Builds directly on existing ML, deep learning, transformer, Python, and AI engineering preparation.
Research continuity	Strong	The parent field and all three research lanes appear across 2025 and 2026 primary venue materials.
Supervisor dependence	Manageable	The field can be pursued through public papers, repositories, models, and benchmarks without specialist hardware or clinical access.
Compute fit	Conditional but workable	Evaluation, PEFT, quantization, pruning, and small-model studies are feasible; full model pretraining is not.
Career transfer	Strong	Skills transfer to multimodal evaluation, model serving, data pipelines, reliability testing, and deployment.
Publication opportunity	Real but competitive	Strong venues publish work in all three lanes, but incremental comparisons are insufficient.

2.3 What Is Frozen and What Remains Open

Frozen	Open for evidence-based selection
Parent field: Multimodal / Vision-Language Learning and Systems	Exact thesis title and research question
Three ranked research fields	Dataset, model family, benchmark, language, and application setting
Public-data and Colab-first constraints	Whether the contribution is a method, evaluation, system, or empirical study
Publication-aware experimental standards	Final venue and submission timing

3. Ranked Top Three Research Fields

These are the only active fields for topic discovery. They are ranked by the combination of research value, execution feasibility, publication potential, compute cost, and compatibility with the researcher's existing preparation.

3.1 Rank 1: Trustworthy and Reliable VLMs

This field studies whether a VLM is grounded in visual evidence, knows when it is uncertain, remains reliable under difficult inputs, and can be evaluated without misleading metrics. It is the primary field because it supports rigorous evaluation-centered work without requiring foundation-model pretraining.

Suitable problem families

- Object, attribute, relation, action, temporal, or textual hallucination.
- Visual grounding and evidence attribution failures.
- Calibration, uncertainty estimation, selective prediction, and abstention.
- Robustness to misleading instructions, irrelevant context, image degradation, or modality conflict.
- Reliability of VLM-as-a-judge and automated evaluation pipelines.
- Multilingual or low-resource evaluation when language coverage is a genuine confound rather than a cosmetic application.

Evidence of active research

CVPR 2025 included PhD, a large-scale visual hallucination evaluation dataset, Critic-V for catching VLM reasoning errors, and ClearSight for hallucination mitigation [4–6]. CVPR 2026 continued with multimodal decoding, visual-expert refinement, and structural probing of hallucination behavior [7–9]. These papers show that evaluation, mitigation, and mechanistic analysis are all active, but they also raise the novelty bar.

Minimum acceptable contribution

- A new and justified failure condition or evaluation protocol, not only another model leaderboard.
- A reproducible measurement study with cross-model or cross-dataset evidence and meaningful error taxonomy.
- A lightweight mitigation with ablation showing why it works and what capability it harms.
- A calibration or abstention method evaluated for both reliability and retained task utility.

Reject or narrow when

- The proposed work only runs several VLMs on an existing benchmark.
- The evaluation depends entirely on one proprietary LLM judge.

- The claimed hallucination metric cannot distinguish perception failure from language-generation style.
- The model cannot complete the benign task reliably enough for safety or robustness metrics to be interpretable.

3.2 Rank 2: VLM Generalization, Adaptation and Distribution Shift

This field studies performance when deployment data differ from training or benchmark data. It is less dependent on model hype because temporal, cross-source, linguistic, visual-style, corruption, and domain shifts remain fundamental deployment problems.

Suitable problem families

- Cross-dataset and cross-source VLM evaluation.
- Domain generalization with few labels or missing modalities.
- Test-time adaptation and continual test-time adaptation.
- Robustness to visual corruptions, style changes, language shifts, or retrieval degradation.
- Generalization of AI-generated image detection across unseen generators, when treated as a visual robustness problem rather than a cybersecurity field.

Evidence of active research

CVPR 2025 included test-time adaptation methods such as Free on the Fly and VLM generalization work such as SCAP [10, 11]. CVPR 2026 continued with multimodal domain generalization using few labels and robust vision-based retrieval-augmented generation under visual degradation [12, 13].

Minimum acceptable contribution

- A deployment-relevant shift with a justified temporal, cross-source, or domain split.
- A strong no-adaptation baseline and comparison with simple recalibration, retraining, or prompting.
- Evidence across multiple shift conditions, not one corruption severity or one target domain.
- A method or analysis that identifies when adaptation helps, fails, or causes catastrophic side effects.

Reject or narrow when

- The work is another small variation on CIFAR-C without a convincing VLM or deployment setting.
- Random splitting destroys the claimed temporal or domain-shift evaluation.
- The proposed method is compared only with a weak source-only baseline.
- The performance gain is within normal run-to-run noise and has no error analysis.

3.3 Rank 3: Efficient and Scalable VLMs

This field studies how to reduce inference cost, memory, latency, or visual-token processing while preserving useful capability. It is conditionally shortlisted because its engineering value is high, but publication becomes difficult when the contribution is only a quantization or pruning comparison.

Suitable problem families

- Visual-token pruning or merging with task-aware selection.
- Adaptive compute, early exit, or dynamic resolution.
- Quantization or compression effects on hallucination, grounding, calibration, and reasoning.
- Small/open VLM capability retention under memory and latency constraints.
- Efficient document or long-context multimodal inference.
- Hardware-aware trade-offs using actual latency, peak memory, throughput, and energy where measurement is available.

Evidence of active research

CVPR 2025 accepted original work on post-training quantization, data-free quantization, and visual-token pruning, including APHQ-ViT, Quantization without Tears, and DivPrune [14–16]. CVPR 2026 continued with collaborative multimode pruning for VLMs [17]. These examples disprove the claim that efficient AI cannot reach strong venues; they also demonstrate that strong publications introduce new methods or formulations and validate them broadly.

Minimum acceptable contribution

- A new selection criterion, optimization objective, allocation mechanism, or reliability-aware compression strategy.
- A careful efficiency-reliability study that reveals a previously unmeasured failure and changes practical model-selection decisions.
- Real systems evidence beyond parameter compression, including latency, memory, throughput, or energy when the claim is deployment efficiency.
- Comparison with a size-matched or compute-matched random baseline when placement or allocation quality is claimed.

Reject or narrow when

- The contribution is only applying INT8, LoRA, or pruning to a new dataset.
- Only accuracy and compression ratio are reported for a systems claim.
- The evaluation uses one architecture, one dataset, or one seed.
- The experiment budget requires repeated full-model training beyond the available compute.

Field	Default verdict	Primary contribution path	Main risk
Trustworthy and Reliable VLMs	Explore first	Evaluation, analysis, or lightweight mitigation	Crowded benchmark-only work
Generalization and Distribution Shift	Explore second	Robust protocol, adaptation, or failure analysis	Artificial or weak shift design
Efficient and Scalable VLMs	Explore conditionally	New method or reliability-aware systems evidence	Incremental compression comparison

4. Preferred Intersections and Scope Boundaries

4.1 Preferred Intersections

The strongest eventual topic may sit at the intersection of the ranked fields. However, the thesis should have one primary claim and at most one secondary lens.

Priority	Intersection	Why it is promising	Caution
A	Reliability × Distribution Shift	Tests whether VLM trustworthiness survives realistic deployment changes. Supports clear protocols and strong failure analysis.	Do not combine too many shift types or trustworthiness metrics.
B	Reliability × Efficiency	Studies whether pruning, quantization, or small-model deployment changes grounding, hallucination, calibration, or reasoning.	Requires both task utility and real efficiency measurements.
C	Generalization × Efficiency	Studies whether an efficient model or inference strategy remains robust outside its benchmark distribution.	Can become experiment-heavy; use one efficiency intervention.

SCOPE RULE Do not combine reliability, adaptation, efficiency, multilinguality, RAG, video, medical data, and a new dataset in one undergraduate thesis. One primary problem plus one controlled secondary lens is the maximum.

4.2 Application Settings That May Be Used

Application settings are test environments, not new parent fields. They may be selected only when data, evaluation, and domain requirements pass the feasibility gate.

- Document and chart understanding.
- Image-text reasoning and visual question answering.
- AI-generated image detection as a generalization or trustworthiness problem.
- Low-resource or Bengali multimodal evaluation, if ground truth can be created or verified responsibly.
- Medical images only with public datasets and without unsupported clinical claims.

- Short-video understanding only if feature extraction and evaluation fit the compute budget.

4.3 Explicitly Parked Fields

The following are not active thesis fields. They may be revisited after the current thesis, but they must not consume literature-review time now.

Parked field	Reason for exclusion from current thesis discovery
Standalone cybersecurity	New domain foundation, no dependable security supervision, and current goals are better served by AI-centered VLM research.
Federated learning	High methodological and systems burden with weaker direct fit to the frozen parent field.
Diffusion or generative-model training	High compute and crowded novelty landscape; may be used only as a tool, not the research field.
3D, robotics, autonomous driving, specialized sensing	Geometry, simulation, hardware, or sensor requirements exceed current scope.
Standalone medical imaging	Requires domain validation that is not reliably available.
Standalone XAI	Evaluation is often weak or subjective; interpretability may be used only when tied to a measurable reliability claim.
Generic RAG or agent application	A working system alone is an engineering project, not a research contribution.
Generic image classification or detection	Highly saturated unless the research claim concerns reliability, shift, efficiency, or another defensible problem.

5. Topic Discovery Protocol

5.1 Funnel

1. **Step 1, define the failure:** State what current VLMs fail to do, under what condition, and why the failure matters.
2. **Step 2, verify the gap:** Check recent primary papers, official repositories, benchmark leaderboards, and subsequent work. A model-generated novelty claim is never sufficient.
3. **Step 3, verify assets:** Confirm dataset access, license, schema, model weights, baseline code, and evaluation tooling.
4. **Step 4, reproduce a baseline:** Run the simplest strong baseline on a small sample before proposing a new method.
5. **Step 5, run a pilot:** Demonstrate that the claimed failure is measurable and that runtime is acceptable.
6. **Step 6, score and challenge:** Apply the scorecard and request a skeptical independent review.
7. **Step 7, freeze one topic:** Complete the Topic Freeze Record and replace field discovery with a versioned Thesis Specification.

5.2 Candidate Limit

Maintain no more than three active topic candidates, one per ranked research field. Investigate one candidate per chat or session. A candidate that lacks a dataset, baseline, evaluation plan, and stated risk is not active; it is only an idea.

5.3 Literature Search Standard

- Begin with one recent primary paper that precisely defines the problem and terminology.
- Trace backward to seminal methods and forward to 2025–2026 competing work.
- Use surveys only for orientation. Verify all decisive claims against original papers or official documentation.
- Record dataset, model, split, baseline, metric, compute, code availability, limitation, and exact relevance for every core paper.
- Do not perform a full literature review until the feasibility gate passes.

6. Mandatory Topic Feasibility Gate

Score each candidate from 1 to 5. A candidate needs at least 25/35, no score below 3 for Dataset Access, Colab Feasibility, or Baselines and Evaluation, and all hard gates must pass.

Criterion	1, weak	3, acceptable	5, strong
Problem clarity	Vague task or solution first	Bounded failure and users	Precise claim, setting, and consequence
Evidence of a real gap	Blog or assumed novelty	Recent papers show limitation	Gap verified across primary sources and baselines
Dataset access and quality	Unavailable or unclear	Public with manageable caveats	Public, documented, licensed, and well matched
Colab Pro feasibility	Requires sustained large compute	Pilot and main runs fit with controls	Comfortable repeated runs with fallback
Baselines and evaluation	Weak or undefined	Reproducible strong baselines and direct metrics	Multiple strong baselines, ablations, robust protocol
Domain-dependence risk	Unavailable expert required	Some learnable domain knowledge	AI/CSE-centered with independently verifiable labels
Career and research value	Narrow demo	Transferable techniques	Strong research and engineering signal

6.1 Hard Gates

- At least one official dataset or benchmark source is accessible and legally usable.
- At least one strong baseline can be reproduced without a paid API.
- A Colab pilot can finish within a practical session and yields interpretable outputs.

- The primary metric directly tests the main claim and cannot be replaced by accuracy alone.
- The contribution can be stated in one sentence without using 'apply', 'combine', or 'use model X on dataset Y' as the main novelty.
- The recent literature search does not show that the same contribution has already been completed.
- A fallback remains meaningful if the proposed method fails.

6.2 Kill Criteria

- Required data, labels, models, or baseline code cannot be accessed or legally used.
- The baseline cannot be reproduced after a bounded debugging effort and no equivalent baseline exists.
- The measured failure disappears under a correct split or stronger baseline.
- The main result depends on one stochastic LLM judge with no validation.
- The method needs sustained GPU resources beyond the available budget.
- The contribution reduces to a leaderboard comparison, application demo, or expected result.

7. Publication-Quality Contribution Standard

A strong thesis and a publishable paper are related but not identical. The thesis must be complete and defensible. Publication additionally requires a contribution that matters beyond the local dataset or implementation.

Contribution type	Acceptable form	Common weak form
Method	New objective, mechanism, architecture component, adaptation rule, or inference strategy with ablations.	Minor loss-term change with marginal gain.
Evaluation or benchmark	New capability definition, validated protocol, reliable annotations, strong baselines, and evidence that rankings or conclusions change.	Collect examples and report model scores.
Empirical measurement	Reveals a systematic failure, boundary, or trade-off across models and conditions with actionable conclusions.	Compare several models and restate known behavior.
Systems or efficiency	New efficiency method or verified accuracy-latency-memory-reliability trade-off on actual hardware.	Report parameter count and compression ratio only.
Dataset	Documented collection, license, representativeness, quality control, benchmark tasks, and reusable hosting.	Small scraped dataset with uncertain rights or labels.
Replication plus extension	Faithful reproduction followed by a substantive new condition, correction, or generalization result.	Reproduce published numbers with no new question.

7.1 Publication Gate

- One primary research question and no more than two supporting questions.

- A contribution statement that names what is new, compared with what, and why it matters.
- At least two to four meaningful baselines, including a simple strong baseline and the closest prior method where feasible.
- Evidence across multiple models, datasets, domains, or shifts appropriate to the claim.
- Ablations that isolate each proposed component from preprocessing, prompt, or training confounds.
- Error analysis and limitations that explain failure cases rather than hiding them.
- Reproducible code, environment, split, prompt, seed, and evaluation details.

VENUE REALITY Efficient AI, trustworthy VLMs, and generalization work all appear in strong computer-vision venues. The field is not the obstacle. Incremental contribution, weak controls, and irreproducible evaluation are the obstacles.

8. Experimental Design Standard

8.1 Research Question Template

Does [intervention or condition] change [primary outcome] for [model and task setting] under [dataset, shift, or constraint], compared with [strong baseline], while preserving [utility, robustness, or efficiency constraint]?

8.2 Baseline Policy

- Lower bound: a simple, transparent baseline or unmodified model.
- Strong practical baseline: a reproducible current method with public code.
- Closest scientific baseline: the method that most directly competes with the proposed contribution.
- Control baseline: random, size-matched, compute-matched, prompt-matched, or no-adaptation control when needed to isolate the claimed mechanism.
- Published numbers may be contextual references, but claims of improvement require locally comparable runs whenever practical.

8.3 Metric Policy

Claim type	Required metric families	Do not rely on
Task capability	Task-specific accuracy/F1/EM/retrieval metrics plus per-category analysis	Aggregate accuracy only
Hallucination/grounding	Capability-specific error rate, grounding score, precision/recall, and human-validated subset	One LLM judge score

Claim type	Required metric families	Do not rely on
Calibration/abstention	ECE or calibration error, risk-coverage curve, AUROC/AUPRC, retained utility	Confidence average
Distribution shift	In-domain versus shifted gap, worst-domain performance, degradation curve, recovery	Random split performance
Efficiency	Latency, throughput, peak memory, model size, compute estimate, and retained quality	Parameter count alone
Robustness/safety	Attack or failure success plus benign utility and false refusal	Failure rate without task utility

8.4 Statistical and Reporting Rules

- Use fixed and documented train/validation/test splits. Use temporal or cross-source splits when the claim concerns shift.
- Report multiple seeds for training-based results when feasible. For expensive inference-only evaluations, use bootstrap confidence intervals or repeated decoding where stochasticity matters.
- Report variance, confidence intervals, or effect sizes when comparing small differences.
- Predefine the primary metric and avoid selecting the best metric after observing results.
- Document prompts, decoding parameters, quantization settings, preprocessing, and judge instructions.
- Treat LLM-as-a-judge as a measurement instrument that requires validation, not as ground truth.
- The NeurIPS checklist is the default rigor checklist for reproducibility, transparency, ethics, and societal impact [18].

9. Compute and Implementation Policy

9.1 Default Compute Strategy

- Prefer open-weight VLMs that can run through quantized inference or adapters on Colab Pro.
- Start with one small or medium model and a small benchmark slice. Expand only after the pipeline and metrics are verified.
- Prefer inference-only evaluation, parameter-efficient adaptation, feature caching, or lightweight heads over full-model training.
- Do not build a thesis that requires training a foundation model, repeated 30B+ model runs, or sustained multi-GPU infrastructure.
- Use stronger temporary hardware only to scale an already validated experiment, never to rescue an infeasible design.
- Record runtime, GPU type, peak memory, precision, batch size, model revision, and software versions for every reported experiment.

9.2 Escalation Rule

A candidate may request limited stronger compute only after it passes dataset access, baseline reproduction, metric validation, and a Colab pilot. The request must estimate the number of runs, expected hours, storage, and the scientific reason the stronger hardware changes the evidence.

9.3 Paid API Policy

- No primary result may depend on a paid proprietary API.
- A proprietary model may appear only as an optional contextual reference when published or free evaluation results are available.
- If API behavior changes over time, record model version, access date, parameters, and any non-reproducibility risk.
- A candidate fails if its central comparison cannot be reproduced without unaffordable API usage.

10. Data, Model, Ethics and Governance

10.1 Dataset Acceptance Checklist

- Official source and stable download location verified.
- License or terms permit the intended academic use and derived reporting.
- Schema, labels, collection process, time period, and known biases understood.
- Duplicates, leakage, benchmark contamination, class imbalance, and source overlap checked.
- Personally identifiable or sensitive data handled only when necessary, lawful, and ethically justified.
- Dataset limitations documented using a datasheet-style record [19].

10.2 Model Acceptance Checklist

- Weights, license, model card, revision, tokenizer/processor, and intended use verified.
- The model can complete the unmodified task well enough for the research metric to be meaningful.
- Training-data contamination and benchmark memorization risks considered where documentation exists.
- Quantization, chat templates, prompt format, and decoding settings are pinned.
- Performance and limitations are reported in a model-card-style summary [20].

10.3 Trustworthiness Scope

Trustworthiness is not a vague synonym for high accuracy. The NIST AI Risk Management Framework identifies valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair characteristics [21]. A thesis should select only the characteristics directly connected to its research question and context of use.

ETHICS RULE Do not overclaim safety, fairness, medical utility, or real-world deployment. Report exactly what the benchmark supports, disclose untested populations and conditions, and state who could be harmed by model errors.

11. Reproducibility and Engineering Standard

11.1 Required Project Artifacts

- README with research question, environment, data acquisition, and exact reproduction commands.
- Version-controlled source code with configuration files rather than notebook-only hidden state.
- Locked environment file, package versions, model revisions, and hardware notes.
- Deterministic or documented split generation and seeds.
- Baseline, proposed method, evaluation, ablation, and figure-generation scripts.
- Experiment registry recording run ID, commit, configuration, data version, result, and failure notes.
- Model and dataset documentation, including licenses and limitations.
- Final tables generated from machine-readable experiment outputs, not manually copied values.

11.2 Suggested Repository Shape

Path	Purpose
configs/	Versioned baseline, method, model, dataset, and ablation configurations.
data/README.md	Acquisition instructions, licenses, hashes, splits, and files that must not be committed.
src/	Reusable data, model, method, and evaluation modules.
scripts/	Explicit preprocessing, training, inference, evaluation, and figure commands.
experiments/	Run manifests and machine-readable summaries, not large checkpoints.
notebooks/	Exploration and presentation only; canonical results must be reproducible through scripts.
paper/	Manuscript, bibliography, figures, tables, and submission checklist.
docs/	Decision log, dataset card, model card, risk register, and experiment protocol.

For dataset or evaluation contributions, current NeurIPS guidance emphasizes accessible code/data, long-term hosting, metadata, limitations, and responsible-AI documentation [22, 23]. These requirements are used as design guidance even when NeurIPS is not the target venue.

12. Thesis-to-Publication Strategy

12.1 Canonical Strategy

Design the thesis from day one so that its data, baselines, protocols, code, ablations, and limitations can support a reputable submission. Do not delay thesis completion to chase venue-specific novelty or endless additional experiments. Once the thesis evidence is complete, decide whether the work is submission-ready or needs a focused extension.

Stage	Thesis objective	Publication-aware action
Topic selection	Choose a finishable, defensible question	Verify the gap and identify the closest competing papers
Baseline	Establish a trustworthy reference result	Use public code, record versions, and reproduce locally
Method/evaluation	Test the central claim	Define contribution and ablation plan before full runs
Full experiments	Produce complete evidence	Use venue-level controls, seeds, error analysis, and limitations
Thesis writing	Explain the work transparently	Maintain paper-ready figures, tables, and bibliography
Post-thesis decision	Close the university deliverable	Submit if the contribution is mature; otherwise add one targeted extension

12.2 Venue Families

Venue selection occurs after the contribution and evidence are known. Computer vision commonly values selective conferences as major publication venues, so a reputable conference is not inferior merely because it is not a journal. Potential venue families include CVPR, ICCV, ECCV, WACV, ACM Multimedia, Transactions on Machine Learning Research, Pattern Recognition, International Journal of Computer Vision, and IEEE Transactions on Pattern Analysis and Machine Intelligence. The final choice must follow the current official scope, quality bar, submission policy, page limit, and dual-submission rules.

12.3 Submission Readiness Gate

- The abstract states one clear contribution and does not rely on inflated novelty language.
- The closest prior work is reproduced or compared under a defensible protocol.
- Results support every title and abstract claim.
- Figures and tables expose trade-offs, variance, and failures, not only best scores.
- Ablations isolate the proposed mechanism.

- Code and data documentation are sufficient for independent reproduction.
- Limitations, ethical considerations, compute, and potential misuse are disclosed.
- The venue is reputable, in scope, and not selected merely because publication appears easy.

13. Research Workflow

13.1 Platform Roles

Platform	Primary role	Do not use it as
Perplexity	Find current papers, official datasets, repositories, benchmark pages, and citation trails.	Final judge of novelty or feasibility.
Claude	Read and synthesize a small set of full papers; compare methods, experiments, and limitations.	Source of unverified current facts.
ChatGPT	Research strategy, feasibility audit, RQ design, experiments, implementation support, and specification updates.	Storage for unrecorded final decisions.
Kimi	Independent skeptical review, stronger baseline search, hidden-complexity and novelty challenge.	Sole evidence source.
Canonical specification	Record the frozen field, accepted topic, rules, decisions, and changes.	A dumping ground for raw papers and temporary notes.

13.2 Required Session Output

- Direct answer in three to eight bullets.
- Evidence table with claim, source, confidence, and limitation.
- Verdict: continue, narrow, park, or reject.
- One next action that can be completed in 30–90 minutes.
- Decision-log entry when direction, scope, dataset, method, or evaluation changes.

13.3 Research Phases Without Calendar Dates

8. **Phase A, orientation:** learn the field vocabulary and read a bounded set of primary papers.
9. **Phase B, candidate screening:** generate no more than three candidates and apply the feasibility gate.
10. **Phase C, pilot:** verify data, baseline, metrics, and compute on a small subset.
11. **Phase D, topic freeze:** complete the Topic Freeze Record and create Thesis Specification v1.
12. **Phase E, baseline system:** build a reproducible end-to-end baseline before implementing the contribution.
13. **Phase F, contribution:** implement one primary intervention or evaluation protocol and its ablations.

14. **Phase G, full evidence:** run controlled experiments, error analysis, robustness checks, and efficiency measurements.
15. **Phase H, thesis and paper:** write the thesis, audit claims, then decide on submission or one focused extension.

14. Operational Templates and Checklists

14.1 Candidate Topic Brief

Field	Required entry
Candidate name	Short descriptive label, not a final title.
Primary research field	Rank 1, 2, or 3 from this specification.
Observed failure	What fails, under which condition, and for whom it matters.
Research question	Testable question with intervention/condition, outcome, setting, and baseline.
Gap evidence	Three to six primary papers and the precise unresolved point.
Dataset/benchmark	Official link, license, size, labels, split, known risks, and access status.
Models	Open weights, license, size, compute estimate, and benign task capability.
Baselines	Simple baseline, strong practical baseline, closest prior method, and necessary controls.
Evaluation	Primary metric, supporting metrics, ablations, error analysis, and success criterion.
Compute	Pilot and full-run estimate, GPU, storage, seeds, and fallback.
Contribution	Method, evaluation, empirical measurement, system, dataset, or replication plus extension.
Kill criteria	Conditions that force rejection or narrowing.

14.2 Topic Freeze Checklist

- Research question is narrow and testable.
- Gap is verified against recent primary work.
- Dataset and model access are confirmed, not assumed.
- A strong baseline has been reproduced.
- A pilot demonstrates the failure or opportunity.
- Compute estimate includes repeated runs and ablations.
- Primary and supporting metrics directly test the claim.
- Contribution survives comparison with a simple strong baseline.
- Risk register and meaningful fallback are written.
- Score is at least 25/35 with no hard-gate failure.

14.3 Strong Security and Trustworthiness Paper Checks

Use this only when the selected VLM topic involves jailbreaks, adversarial robustness, privacy, harmful content, or security. Security is then a property of the VLM system, not a reopening of standalone cybersecurity.

- Define the protected property, attacker capability, attack surface, and success condition.
- Measure benign task utility alongside attack or failure success.
- Use a baseline attack, baseline defense, and no-attack control.
- Separate inability to perform the task from resistance to the attack.
- Report false refusal, transferability, stochasticity, and prompt/model version.
- Do not release harmful artifacts without a proportionate risk review.

14.4 Paper Writing Checklist

- Title and abstract match the actual evidence and scope.
- Introduction identifies the problem, gap, contribution, and evidence without unsupported market claims.
- Related work compares mechanisms and limitations, not one-paragraph summaries.
- Method is precise enough to reproduce, including prompts and preprocessing.
- Experimental setup names data, splits, models, baselines, compute, seeds, and metrics.
- Results answer each research question and include uncertainty where relevant.
- Ablations isolate every claimed component.
- Error analysis explains representative successes and failures.
- Limitations distinguish untested from failed conditions.
- Code, configuration, data instructions, and references are complete.

15. Topic Freeze Record

Complete this section only after a candidate passes the feasibility gate. When completed, create a separate versioned Thesis Specification v1. Do not overwrite this field-level specification.

Specification item	To be completed after topic freeze
Working title	
Primary research field	
Research question	
Problem setting and boundaries	
Dataset and access status	
Models and licenses	
Proposed contribution	
Explicit non-contributions	
Reproducible baselines	
Metrics, ablations, and success criteria	
Compute budget and fallback	
Core papers	

Specification item	To be completed after topic freeze
Main risks and kill criteria	

16. Decision Log

Date	Decision	Status	Evidence-based reason	Next action
20 Aug 2026	Broad thesis exploration	Closed	Multiple fields were under consideration; no topic was frozen.	Apply a common feasibility framework.
21 Aug 2026	Cybersecurity field screening	Parked	Standalone cybersecurity added domain and supervision risk; reviewed candidates also contained dataset, baseline, or evaluation problems.	Retain security only as a future field or VLM property.
22 Aug 2026	Parent field freeze	Active	Multimodal/VLM research offers the best balance of background fit, active research, public assets, AI-career transfer, and self-directed feasibility.	Screen Rank 1 candidate topics first.
22 Aug 2026	Three ranked research fields	Active	Trustworthiness first, distribution shift second, efficiency conditional third.	Produce field-level feasibility cards before topic generation.

16.1 Future Decision Entry

Date	Candidate/change	Verdict	Evidence and limitation	Next action
		Continue / Narrow / Park / Reject		
		Continue / Narrow / Park / Reject		
		Continue / Narrow / Park / Reject		

17. Change Control

- **Patch update, v1.0.x:** wording, source links, minor clarification, or formatting with no decision change.
- **Minor update, v1.x:** research-lane ranking, feasibility threshold, or workflow change supported by new evidence.

- **Major update, v2.0:** parent-field change. Requires documented failure of the current field, a replacement feasibility audit, and explicit decision log.
- **Topic specification:** created as a separate document after topic freeze; it does not replace or silently rewrite this field specification.

References and Primary Sources

[1] IEEE/CVF, CVPR 2025 Call for Papers.

[2] IEEE/CVF, CVPR 2026 Call for Papers.

[3] IEEE/CVF, CVPR 2026 Tentative Schedule.

[4] Liu et al., PhD: A ChatGPT-Prompted Visual Hallucination Evaluation Dataset, CVPR 2025.

[5] Zhang et al., Critic-V: VLM Critics Help Catch VLM Errors in Multimodal Reasoning, CVPR 2025.

[6] Yin et al., ClearSight: Visual Signal Enhancement for Object Hallucination Mitigation in Multimodal LLMs, CVPR 2025.

[7] Jiang et al., Breaking the Illusion: When Positive Meets Negative in Multimodal Decoding, CVPR 2026.

[8] Jeong et al., Draft and Refine with Visual Experts, CVPR 2026.

[9] He et al., Structural Graph Probing of Vision-Language Models, CVPR 2026.

[10] Dai et al., Free on the Fly: Enhancing Flexibility in Test-Time Adaptation with Online EM, CVPR 2025.

[11] Zhang et al., SCAP: Transductive Test-Time Adaptation via Supportive Clique-based Attribute Prompting, CVPR 2025.

[12] Li et al., Towards Multimodal Domain Generalization with Few Labels, CVPR 2026.

[13] Chen et al., RobustVisRAG: Causality-Aware Vision-Based Retrieval-Augmented Generation under Visual Degradations, CVPR 2026.

[14] Wu et al., APHQ-ViT: Post-Training Quantization with Average Perturbation Hessian Based Reconstruction for Vision Transformers, CVPR 2025.

[15] Fu et al., Quantization without Tears, CVPR 2025.

[16] Alvar et al., DivPrune: Diversity-based Visual Token Pruning for Large Multimodal Models, CVPR 2025.

[17] Wu et al., Collaborative Multi-Mode Pruning for Vision-Language Models, CVPR 2026.

[18] NeurIPS, Paper Checklist Guidelines.

[19] Gebru et al., Datasheets for Datasets, Communications of the ACM, 2021.

[20] Mitchell et al., Model Cards for Model Reporting, FAT* 2019.

[21] NIST, Artificial Intelligence Risk Management Framework 1.0.

[22] NeurIPS 2026, Evaluations and Datasets Track Call for Papers.

[23] NeurIPS 2026, Code and Data Submission Policy.

Source note: The Paper Digest CVPR 2025 topic clustering was used only as an orientation aid. Field decisions and technical claims in this specification rely on official venue pages, primary papers, or recognized standards.